ITAI 1371 – Module 12

Individual Contribution Submission

Alexander DeBusk

Ethics, Fairness, and Bias in Machine Learning

L12 – Contributions – Alexander DeBusk

- Reviewed and completed the lab activities involving machine learning ethics, fairness, and bias.
- Analyzed the model's accuracy for male and female subgroups and compared the differences in performance.
- Calculated and interpreted the False Positive Rates (FPR) and False Negative Rates (FNR) for both groups.
- Completed the reflective knowledge check, explaining the potential consequences of bias in loan approvals and hiring decisions.
- Examined how proxy bias can remain even after removing a sensitive feature such as sex.
- Completed the reflection journal discussing what I learned about fairness measurements and bias mitigation.
- Correctly saved all deliverables as PDFs using the appropriate file-naming conventions and submitted them to the group leader on time.

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Individual Contribution Submission

Shareefah Adeola

Ethics, Fairness, and Bias in Machine Learning

CONTRIBUTION JOURNAL: Module 12: Lab Title: Ethics, Fairness, and Bias in Machine Learning

ITAI 1371

Student: Shareefah Adeola

Date: November 12, 2026

1. Overview of My Contribution

I completed the Module 12 lab on Ethics, Fairness, and Bias in Machine Learning. My work involved training and evaluating a Logistic Regression model using the Adult Income dataset and examining whether the model produced different results across male and female demographic groups.

2. Tasks Completed

- Executed the Python code to load and preprocess the Adult Income dataset.
- Trained the Logistic Regression model and obtained an overall accuracy of 84.61%.
- Evaluated model accuracy separately for males and females.
- Calculated and compared False Positive Rates (FPR) and False Negative Rates (FNR).
- Completed the reflective knowledge check and evaluated the ethical implications of the model's predictions.

3. Results and Findings

Performance Metric	Males	Females
Accuracy	81.20%	91.81%
False Positive Rate	10.26%	2.81%
False Negative Rate	37.80%	47.84%

The results revealed differences in model performance and error rates between the two demographic groups, indicating potential fairness concerns.

4. Challenges and Learning

One important lesson was that overall model accuracy does not necessarily guarantee fairness. I learned that a model can perform well overall while producing unequal outcomes for different demographic groups.

I also gained practical experience interpreting confusion matrix results and understanding the ethical implications of false positive and false negative predictions.

5. Conclusion

This lab improved my understanding of algorithmic bias, fairness assessment, and responsible machine learning. I learned the importance of evaluating models across demographic groups and applying appropriate bias mitigation techniques before deploying machine learning systems in real-world situations.

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Individual Contribution Submission

Yilin Leng

Ethics, Fairness, and Bias in Machine Learning

Contribution

In this Module 12 Lab, I worked on a fairness audit of a machine learning model using the UCI Adult dataset. In Google Colab, I loaded and prepared the dataset, including removing missing values with dropna. I also changed the income target variable into binary values. Then, I built and tested a machine learning pipeline using StandardScaler, OneHotEncoder, and LogisticRegression. The model achieved an overall accuracy of 84.61%. For the subgroup analysis, I implemented and tested the get_subgroup_accuracy() function. I used Boolean masking to separate male and female samples from the test dataset and calculated the accuracy for each group. The results showed that male accuracy was 81.20% and female accuracy was 91.81%, with a difference of 10.61 percentage points. I compared these results to understand how the model performed for different groups and checked that the calculations were correct. I also implemented the get_rates() function and used confusion_matrix from sklearn.metrics to calculate TN, FP, FN, and TP. Then, I calculated the FPR and FNR for both groups. The results showed that males had an FPR of 10.26% and an FNR of 37.80%, while females had an FPR of 2.81% and an FNR of 47.84%. I compared these results to better understand the prediction errors for each group. After completing the technical parts of the lab, I also completed the Reflective Knowledge Check and ethical analysis. I used the experiment results to explain Proxy Bias and how features such as occupation, education, hours worked per week, and marital status may still be related to gender. I also discussed ways to reduce bias, such as reweighting and fairness constraints. In addition, I explained how incorrect predictions could affect people in real situations, such as job hiring and loan applications. During this lab, I checked the code outputs, reviewed the accuracy and error rates for both groups, and helped organize the results and

written explanations. I made sure the data was consistent and clearly presented in the final submission. Through this lab, I gained more experience with data preparation, model training, and fairness analysis. I also learned that when using machine learning in real life, we need to consider fairness and ethical issues instead of focusing only on accuracy.